



Introduction to Modelling and Simulation

What You Will Complete Today

- Define a system, model, and simulation in your own words.
- Explain when simulation is useful and when it is unnecessary.
- Identify entities, resources, state variables, events, inputs, and outputs.
- Frame one real system for later modelling work.

Simulation Evaluates a Model Numerically

- **System:** interacting entities working toward a purpose.
- **Model:** assumptions that describe how the system works.
- **Simulation:** computer-based experiments performed on the model.
- The result is evidence about the model, not direct experimentation on the real system.

Use the Core Terms Precisely

- **Entity:** an object represented in the system.
- **Attribute:** a value describing an entity.
- **State variable:** information needed to describe the system now.
- **Event:** an occurrence that can change system state.
- **Performance measure:** an output used to judge alternatives.

Use Simulation When Direct Study Is Impractical

- The real system is costly, risky, disruptive, or unavailable.
- The system is too complex for a useful analytic solution.
- Several designs or operating policies must be compared.
- Behavior must be observed over compressed or expanded time.
- Internal system states are difficult to measure directly.

Simulation Is Not Always the Right Tool

- Use an analytic solution when it answers the question accurately and efficiently.
- Do not simulate when reliable input data cannot be obtained.
- Do not simulate when direct experimentation is cheaper and safe.
- Do not begin without enough time, resources, and a clear decision question.
- A simulation cannot repair an ill-defined problem.

Choose the Least Disruptive Study Method

- **Actual system:** experiment directly when practical and safe.
- **Physical model:** build a scaled or tangible representation.
- **Analytic model:** solve assumptions through mathematics.
- **Simulation model:** evaluate assumptions through numerical experiments.
- The study objective determines which method is appropriate.

A System Has Interacting Parts

- **Entities:** customers, packets, patients, jobs, or products.
- **Resources:** servers, nurses, machines, vehicles, or channels.
- **State:** queue length, resource status, inventory, or location.
- **Events:** arrivals, departures, failures, repairs, or completions.
- **Measures:** delay, utilization, throughput, cost, or service level.

A Model Is a Purposeful Simplification

- A model keeps only the details needed to answer a defined question.
- The same real system can have different valid models for different purposes.
- Every omitted detail is a modelling choice that should be explainable.
- More detail can increase cost without improving the decision.
- Model quality means fitness for purpose, not maximum complexity.

Define the Boundary Before Building

- **Inside:** components represented explicitly.
- **Outside:** environmental factors treated as inputs or assumptions.
- **Time horizon:** how long the system is observed.
- **Level of detail:** the smallest behavior that matters to the question.
- A clear boundary prevents uncontrolled scope growth.

Document Assumptions Before Programming

- State each simplifying assumption clearly.
- Record its reason, evidence source, and expected effect.
- Separate confirmed facts from provisional choices.
- Review assumptions with people who understand the real process.
- Retain limitations and unresolved questions for later validation.

Identify Inputs, Logic, State, and Outputs

- **Inputs:** arrival rates, service times, capacities, or policies.
- **Logic:** rules that transform inputs into system behavior.
- **State:** values updated while the model runs.
- **Outputs:** measures collected for comparison and decisions.
- Units and definitions must remain consistent throughout the study.

Example: A Clinic Queue

- Entities: arriving patients.
- Resources: nurses and consultation rooms.
- Events: patient arrival, service start, and service completion.
- State: patients waiting and staff availability.
- Measures: waiting time, queue length, utilization, and patients served.

Start with a Decision Question

- **Weak:** How does the clinic work?
- **Stronger:** How many nurses keep median waiting below 20 minutes?
- Name the decision maker and the decision they control.
- Choose measures that answer the question directly.
- Set the scenarios that must be compared.

Plan Runs, Scenarios, and Replications

- **Run:** one execution under stated inputs and assumptions.
- **Scenario:** one coherent alternative to test.
- **Replication:** an independent repeat used to measure uncertainty.
- **Experiment:** a planned comparison of scenarios and replications.
- Stochastic models require multiple runs; one result is not the answer.

Simulation Supports Safe What-If Analysis

- Compare alternatives without disrupting the current system.
- Estimate performance under projected operating conditions.
- Control experimental inputs more tightly than in the real world.
- Study slow or rapid processes using a flexible simulation clock.
- Observe rare or risky conditions without exposing people or assets.

Simulation Produces Estimates, Not Exact Answers

- Model development can require substantial time, data, and expertise.
- Outputs inherit uncertainty from random inputs and assumptions.
- Animation can look convincing even when the model is wrong.
- Optimization and validation can be difficult.
- Credible conclusions require statistical analysis and transparent limitations.

Avoid the Most Common Pitfalls

- No clear objectives or decision question.
- Too much or too little model detail.
- Poor data collection and undocumented assumptions.
- Treating simulation as programming only.
- Using a single run, weak statistics, or output animation as proof.

Verification and Validation Ask Different Questions

- **Verification:** Did we implement the model correctly?
- **Validation:** Does the model represent the real system well enough?
- Verification uses debugging, traces, tests, and code reviews.
- Validation uses data comparison, sensitivity checks, and expert review.
- Both will be developed further later in the course.

Guided Practice: Map a Checkout System

- Identify the entities, resources, state variables, and events.
- Write one decision question about staffing or queue performance.
- Draw the system boundary and list three assumptions.
- Choose two inputs and two performance measures.
- Explain why simulation may add value.

Discuss the Modelling Choices

- Which details belong inside the checkout model boundary?
- When would direct observation be better than simulation?
- Which assumption is most likely to change the conclusion?
- What data would you request before building the model?
- How would you explain the model's limitations to a decision maker?

Key Takeaways

- Simulation evaluates a simplified model of a real system.
- The study begins with a decision question, boundary, assumptions, and measures.
- Simulation is powerful when direct or analytic study is impractical.
- Results are estimates whose credibility depends on data, logic, and analysis.
- Clear documentation is part of the model, not an optional extra.

Week 1 Checkpoint and Next Step

- Submit a one-page map of one real system discussed today.
- Include the question, boundary, entities, resources, state, events, inputs, and outputs.
- This is a readiness checkpoint, not Laboratory Exercise 1.
- Next week: classify models and organize a complete simulation study.

Thank you!